1. A.)What do mean by batch processing?

   B.) Describe the terms

   (i) Job Control Language

   (ii) Batch Monitor

2. Explain and differentiate between single processor, multiprocessor systems and client server systems, highlighting their advantages and disadvantages if any.

3 Answer the following (5 marks each)

   a. Briefly explain the two types of multiprocessor system configurations.

   b. Compare and contrast the terms Process and Program.

4. (5marks each)

   a. What is meant by context switch and describe how it takes place.

   b. Explain the terms: (i) multiprogramming (ii) multitasking

5. What is meant by a system call? How does it work? How is it different from a normal function call? Also explain fork, exec and wait system calls.

6. Define an O.S. and provide an abstract view of the same. Also list and explain the responsibilities of a general purpose operating system.

7. Distinguish between Time-sharing system and Real Time systems.

8. Describe any 5 process Related system calls and File related System calls.

9. Explain the terms

   a. Competition

   b. Cooperation

   With respect to Concurrent processes.

10. Explain the following term. Also indicate their difference and their use.

    (i) long term scheduler (ii) medium term scheduler (iii) short term scheduler

11. What is meant by a process control block? Explain. Also with a neat diagram explain the different states of a process.

12. What is meant by multithreading? List and explain the 3 multithreading models with neat diagrams.

13. (5 marks each)

    (i) Explain the role of time slice on the performance in scheduling algorithms.

    (ii) Explain the difference between a process and a thread.

14.* (5 marks each).

    (i) What is meant by Belady's anamoly?

(ii) What is meant by preemption? What is its use?

15. Describe the term Context Switch with a Diagram illustrating the same.

16. List and explain the 5 criteria used to compare the various CPU scheduling algorithms.

17. Four jobs arrive at a computer at times 0,1,2,3 respectively. They have the estimated running times of 9,5,4 and 8 seconds. Their priorities are 3, 4, 5, and 2 respectively, with 5 being the highest priority. Draw *Gantt charts* and determine the *turnaround-time* and *waiting time* for each process in the following cases:

(i) Round robin with time slice = 3 time units

(ii) Shortest job first with preemption.

(iii). Priority scheduling with preemption.

18. With neat diagrams explain the scheduling algorithms

(i) Round Robin Scheduling

(i) Multilevel Queue Scheduling with feedback.

19. Four jobs arrive at a computer at almost the same time. They have the estimated running times of 12, 5, 4, and 3 minutes. Their priorities are 3, 4, 5 and 2 respectively, with 5 being the highest priority. Consider the process switching overhead = 1 time unit. Draw *Gantt charts* and determine the *turnaround-time* and *waiting time* for each process in the following cases:

(i) Round robin with time slice = 3 time units

(ii) Shortest job first.

(iii). Priority scheduling.

20. Assume that the following five processes arrive at time 0,1,2,3 and 4 respectively, with the length of the CPU burst time given in milliseconds:

Process	CPU Burst Time
P1	6
P2	13
P3	7
P4	5
P5	15

Consider FCFS, SJF with preemption and RR (quantum=10) scheduling algorithms and determine the average turnaround time and average waiting time in each case, with the help of Gantt charts.

21. diffentiate between preemptive and non-preemptive scheduling. Describe advantages and Disadvantages of these two scheduling schemes.

22. Write a C code for the following.

a.) use fork to create a child process to execute shell using exec system call.

b.) Create a thread which executes a shell with exec system call.

23.* How can we design a graph for depicting the precedence relationship in a Multitasking Scenario. Illustrate with an example.

24.. How can multitasking improve the efficiency of a system? Give an example which demonstrate that multitasking increases the efficiency of the CPU.

25. List the concurrency related problems. How can they be eliminated?

26. *Illustrate the use of multiple buffering scheme to improve the efficiency of a multitasking system.

27. What is thread safe programs? What are the precautions should be taken while designing multithreaded programs?

28. With adequate examples explain the terms:-
(i) critical section (ii) race condition

29. What is meant by a race condition? Explain with the help of an example. Also list and explain the conditions which should be satisfied by any solution to a critical section problem.

30. What do you understand by message passing? Explain this scheme under (i) direct communication and (2) indirect communication.

31. Explain the use of shared memory for inter process communication. Describe the system calls for allocating and attaching shared memory segments.

32. Explain the two models of Inter Process Communication and the benefits of IPC.

33. What are the hardware support available for implementing semaphore?

34. Describe the pessimistic and optimistic concurrency control techniques?

35. Describe the Software approach to solve the critical section problem. Illustrate how this approach cannot solve the problem without special hardware support?

36. Comment on the Queuing implementation of semaphores.

37. Explain the use of the following instructions for mutual exclusion.
(i) TestAndSet instruction and (ii) Swap instruction

38. What is a semaphore? What are the 2 operations associated with it? Explain them. Also explain how they could be used to achieve synchronization with the help of an example.

39. Describe the implementation of critical regions. How are they better than the usage of semaphore?

40. Describe the implementation of Monitors.

41. Describe the implementation of Wait and signal using monitors.

42. Write the structure of the producer process as well as the consumer process in the classical solution to "Bounded buffer" critical section problem. Clearly indicate (i) the number of semaphores used (ii) their initial values (iii) purpose of using them.

43. Solve the Bounded Buffer producer/consumer problem using monitors.

44. How the message passing be used for mutual exclusion. Discuss the implementation.

45. Write the classical solution for First Reader writer critical section problem. Clearly indicate (i) the number of semaphores used (ii) their initial values (iii) purpose of using them.
46. Solve the bounded buffer producer/consumer problem using messages.
47. *State the classic "Dining philosophers problem" and provide a solution for the same using semaphores, giving adequate comments or explanation. Clearly indicate (i) the number of semaphores used (ii) their initial values (iii) purpose of using them. When will this solution lead to a deadlock and how can the deadlock be avoided?

48. Explain the following terms (i) deadlock prevention (ii) resource allocation graph and wait for graph.
49. Describe the algorithm for Deadlock avoidance. Illustrate it with an example.
50. What is meant by a deadlock? List and explain the conditions that cause a deadlock.
51. What is the difference between deadlock prevention and deadlock avoidance? How can you achieve deadlock prevention? Explain.
52. Describe the banker's algorithm for detection and recovery. Illustrate it with an example.
53. What is meant by deadlock recovery? Explain in detail the 2 ways of recovering from a deadlock.
54. Consider the following snapshot of a system

	Allocation	MAX	Available
	A B C	A B C	A B C
P0	0 0 1	0 0 1	1 5 2
P1	1 0 0	1 7 5	
P2	1 3 5	2 3 5	
P3	0 6 3	0 6 5	
P4	0 0 1	0 5 5	

A, B and C are the resource types. P0, P1, P2, P3 and P4 are the 5 processes.
The current allocation, the maximum resources required by each process and the available resources have been given. Answer the following questions using Bankers algorithm.

(i) Determine the maximum number of resources of each type in the system.
(ii) Determine the Need matrix.
(iii) List the steps in determining whether the system is safe or not. Give the safe sequence if present.
(iv) If a request from P4 arrives for (0,4,3) can the request be granted immediately?

55. Explain the following terms (i) external fragmentation (ii) internal fragmentation (iii) compaction.
56. What is meant by paging? Draw a diagram which indicates the hardware support for paging and explain how logical address is converted to a physical address in this scheme.
57. Draw the diagram of paging hardware with TLB and explain the significance of TLB.
58. Explain the concept of memory management through various types of memory partitioning schemes.
59. Write short notes on (i) memory protection in a paged environment (ii) shared pages.

60 Explain the basic concept of memory segmentation with a neat block diagram.

60. Discuss the following with an adequate example each.
    a. First come First Served based page replacement policy
    b. Least Recently used page replacement policy
    c. Optimal page replacement policy

61. Consider a page size of 100 bytes and the following memory address reference string: 120, 220,312, 423, 211, 115, 543, 653, 234, 167, 278, 190, 225, 321, 765, 666, 333, 222, 111, 249, 339, 666

    How many page faults would occur for the following replacement algorithms, assuming 4 page frames?
    a. LRU replacement
    b. FIFO replacement
    c. Optimal replacement.

62. What is meant by a page fault? Explain briefly. Consider the following page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6
    How many page faults would occur for the following replacement algorithms, assuming 3 page frames?
    a. LRU replacement
    b. FIFO replacement
    c. Optimal replacement

63. What is meant by a page fault? Explain briefly. Consider the following page reference string: 5, 7, 8, 9, 7, 6, 10, 11, 7, 6, 7, 6, 7, 8, 12, 11, 8, 7, 6, 7, 8, 11
    How many page faults would occur for the following replacement algorithms, assuming 3 page frames?
    a. LRU replacement
    b. FIFO replacement
    c. C. Optimal replacement

64. Consider a swapping system in which memory consists of the following partitions in the following order: 10k, 4K, 20K, 18K, 7K, 9K, 12K, and 15K. Which partitions are taken for successive memory requests of 12K, 10K, 9K, 4K for the best fit, first fit and worst fit policies. In this case which algorithm makes most efficient use of memory?

65. Given memory partitions of 100K, 500K, 200K, 300K, 150K and 600K (in order), how would each of the First-fit, Best-fit and worst fit algorithms place the memory segments of processes of 156K, 120K 212K, 417K, 112K and 426K (in order)? In this case which algorithm makes most efficient use of memory?

66. Given memory partitions of 200K, 600K, 300K, 400K, 250K and 700K (in order), how would each of the First-fit, Best-fit and worst fit algorithms place the memory segments of processes of 256K, 220K 312K, 517K, 212K and 526K (in order)? In this case which algorithm makes most efficient use of memory?

67. What is meant by virtual memory and a page fault? When does a page fault occur? Describe the actions taken by the operating system when a page fault occurs.

68. Explain the concept of virtual memory and demand paging.

69. Describe the Cyclic executive approach to schedule real time process, mention the advantages and disadvantage.
70. Illustrate the usage of Time line and Gannt chart for scheduling real time jobs.

71. Consider 2 processes P1 and P2. The periods for P1 and P2 are 50 and 100 respectively. And their processing times are 20 and 35 respectively.
    (i) Is it possible to schedule these tasks based on CPU utilization test?
    (ii) Draw the Gantt chart which depicts the Rate Monotonic scheduling for the above 2 processes. Do the processes meet their deadlines?

72. Consider 2 processes P1 and P2. The periods for P1 and P2 are 50 and 80 respectively. And their processing times are 25 and 35 respectively.
    (i) Is it possible to schedule these tasks based on CPU utilization test?
    (ii) Draw the Gantt chart which depicts the Rate Monotonic scheduling for the above 2 processes. Do the processes meet their deadlines in this case?

73. Describe (i)Rate monotonic Scheduling (ii)Deadline Monotonic scheduling.
74. Comment of the appropriate usage of Utilization Based -Test for analyzing the scheduling of a set of real time processes?
75. How we can solve the blocking problem is real time systems?
76. Consider 3 tasks as shown in table. Find the utilization factor of each task. Can the tasks be scheduled? Give reason for your answer. Also draw the timeline chart for the same. Higher number indicates higher priority and vice versa.

	Period T	Comp. Time, C	Priority, P
Task1	80	40	1
Task2	40	10	2
Task3	20	5	3

77. Consider 3 tasks as shown in table. Find the utilization factor of each task. Can the tasks be scheduled? Give reason for your answer. Also draw the timeline chart for the same. Higher number indicates higher priority and vice versa.

	Period T	Comp. Time, C	Priority, P
Task1	50	12	1
Task2	40	10	2
Task3	30	10	3

78. Compute the response times for the following tasks based on deadline rate monotonic algorithm. Can the processes be scheduled? Also draw the Gantt chart for the same. Note:- Higher value indicates higher priority.

	Period T	comp. Time, C	Priority, P
Task1	7	3	3
Task2	12	3	2
Task3	20	5	1

79. Comment on priority inversion in real time systems. Illustrate with an example.
80. Describe the priority ceiling protocols (OCPP and ICPP).
81. List the essential properties of the following types of operating systems:
    a. Batch
    b. Interactive

c. Real time
d. Distributed

82. A. Describe the need for process control block to manage the processes.
B. Which is the best data structure to implement the process control block? Justify your answer.
83. Write technical notes on
Indefinite postponement
Deadlock prevention
Hardware solutions for achieving mutual exclusion
84. Explain at least one technique to avoid deadlocks.
85. Explain the concept of monitor.
86. What is virtual memory management? How is it realized in traditional operating systems?
87. Bring out the differences among the FIFO and LRU page-replacement policies. Give examples to justify your answers.
88. Describe the various protection mechanisms of segmentation allocation policy.
89. Comment on temporal scope.
90. *How can a real time system be made fault-tolerant in terms of time? Comment on forward recovery.
91. What is rate monotonic scheduling? Give an example.
92. Prove that the Deadline monotonic priority ordering is optimal
93. Write a note on process interaction and blocking.
94. Draw the time line for the following execution sequences of four processes.

Processes	Priority	Execution sequence	Release time
P1	4	EEQQVE	4
P2	3	EQVE	3
P3	2	EVE	2
P4	1	EQQVVE	1

Where Q and V are critical section codes.

95. Draw the time line for the following execution sequences of four processes.

Processes	Priority	Execution sequence	Release time
P1	4	EEQQE	3
P2	3	EVE	3
P3	2	EVE	1
P4	1	EQQVE	0

Where Q and V are critical section codes.

96. Draw the time line for the following execution sequences of four processes.

Processes	Priority	Execution sequence	Release time
P1	2	EEQQVE	2
P2	4	EQVE	3
P3	1	EVE	2
P4	2	EQVVE	0

Where Q and V are critical section codes.

97. Describe Earliest Deadline First scheduling algorithm with an example.
98. Describe Fixed priority scheduling algorithm based on response time analysis method.
99. Describe the method of reducing the blocking time of higher priority processes with an example.
100. Write a note on task management of real time operating system.
101. What are the features of operating systems for IOT
102. Compare the various disk scheduling algorithms.

103. Distinguish between monolithic and micro kernels.
104. Comment on event driven programming